

Project Title: AI Data Scientist Agent

Project Motivation ("the problem"): Retail and e-commerce companies generate massive amounts of data from customer transactions, browsing behavior, inventory, and marketing campaigns. However, most organizations struggle to turn this raw data into actionable insights. Data scientists spend over 40% of their time cleaning and preparing messy data, leaving less time for analysis and decision-making. Existing tools assume data is clean and focus mainly on model training, which overlooks the hardest and most time-consuming parts of the workflow. This leads to delays, missed opportunities, and inefficient operations. Our motivation is to solve these challenges by stepping into the shoes of "Sara, the data scientist," and designing a system that directly addresses her pain points and enables her to generate actionable and deep insights.

Expected Outcome ("the solution"): The project will deliver an AI Data Scientist Agent: a multi-agent system that automates the end-to-end data science workflow for retail and e-commerce. The system will include agents for automated data cleaning, hypothesis testing, feature engineering, model training, and actionable insight generation. The expected outcome is a prototype platform that can reduce analysis time, improve the accuracy of insights, and empower decision-makers with clear, actionable recommendations. The solution will demonstrate measurable improvements in efficiency, enables to generate actionable / deep insights, making recommendation, data visualization, explaining "why" and creating business impact across different functions such as inventory planning, digital marketing, SEO etc. . AI agent designed to integrate with existing retail data sources working through MCP, API integration etc.

Deliverable 1: The first deliverable will focus on project planning and system design. This includes a literature review of automated data science and multi-agent systems, analysis of retail data characteristics, and identification of system requirements. A high-level architecture of the AI Data Scientist Agent will be developed, defining the roles of each sub-agent and the overall workflow. Sample datasets will be explored, and tools, technologies, and evaluation metrics will be finalized.

Deliverable 2: The second deliverable will focus on developing and testing the core sub-agents that perform essential data science tasks. These will include cleaning raw retail datasets by handling missing values and inconsistencies, performing statistical hypothesis tests such as t-tests or chi-square, and validating their outputs on small sample datasets. The goal is to ensure that each sub-agent can function independently and produce reliable results before integration.

Deliverable 3: The third deliverable will involve integrating all the sub-agents into a single working prototype. At this stage, the agents will be connected in a sequence to automate the workflow from data cleaning to insight generation. The system will also incorporate basic feature engineering and predictive modeling capabilities, enabling it to produce not only descriptive but also forward-looking insights. This version of the system will be capable of generating plain-language summaries and recommendations that mimic how a data scientist would communicate findings.

Deliverable 4: The fourth deliverable will be the evaluation and case study phase, where the integrated system will be tested on a full retail dataset. The performance of the AI Data Scientist Agent will be compared against traditional manual analysis in terms of speed, accuracy, and quality of insights. This phase will also include a case study, such as predicting customer churn or identifying top-selling product categories, to demonstrate the system's practical value. The final report will summarize the results, limitations, and potential improvements for future development.

Deliverable 5: If time permits, the final deliverable will be the creation of a simple user interface or dashboard that allows users to upload a dataset and view the automatically generated insights in an intuitive, visual format. This will make the prototype more interactive and demonstrate how the tool could be used in real business environments.